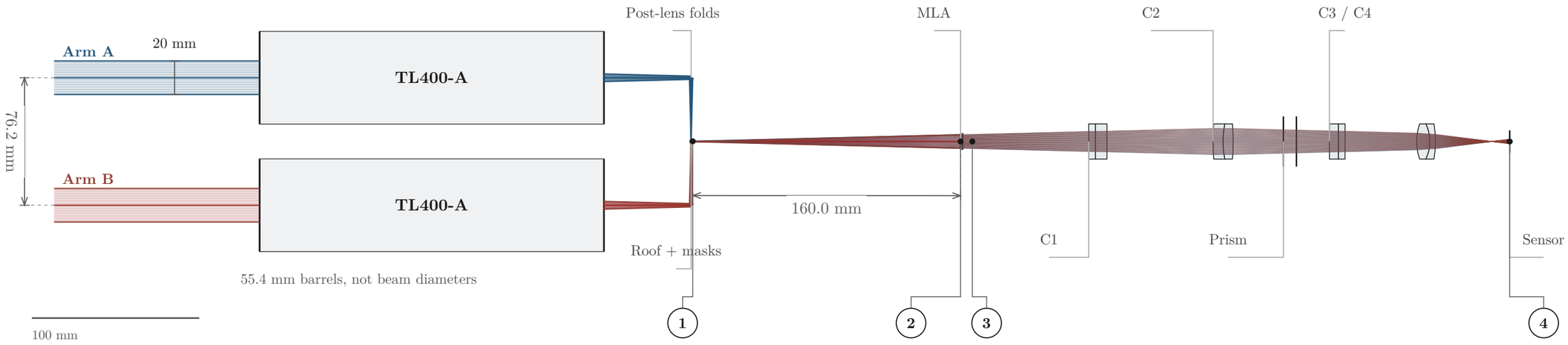
Optical layout and propagated intensity

One two-beam camera module: real R07 anamorphic optics and native turbulent fields. Numbered markers identify the image planes below.

Full top view: X-Z

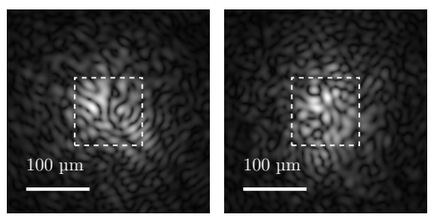
One physical scale | arm A: blue; arm B: red | geometric rays at 400 nm



Native R07 intensity with independent A/B turbulence

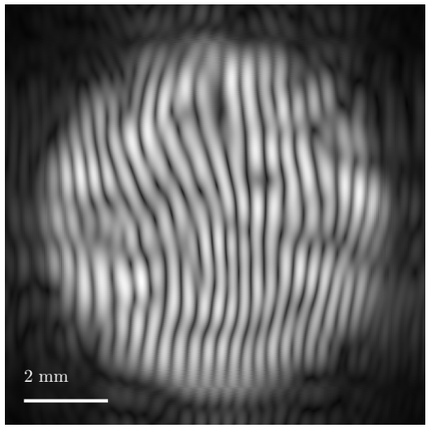
Planes 1-3: 500 nm | plane 4: five wavelengths | enlarged insets, each with its own scale bar

(1) Focal masks A / B



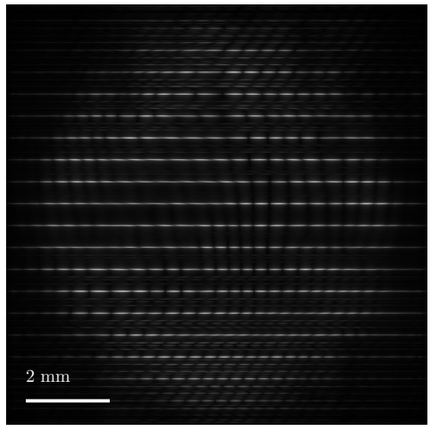
Unmasked focal images; dashed squares: 106.7 µm (1.1 arcsec) masks. Shared A/B contrast.

(2) Overlapped pupil



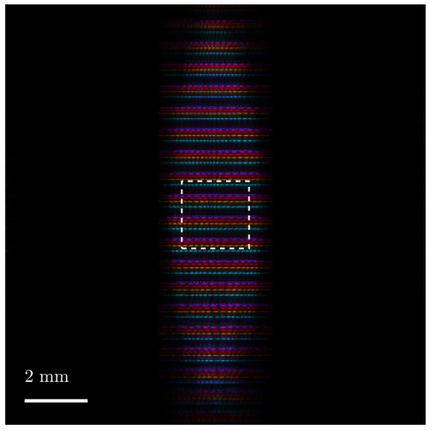
Before the cylindrical array; 8 mm nominal pupil.

(3) Lenslet line foci



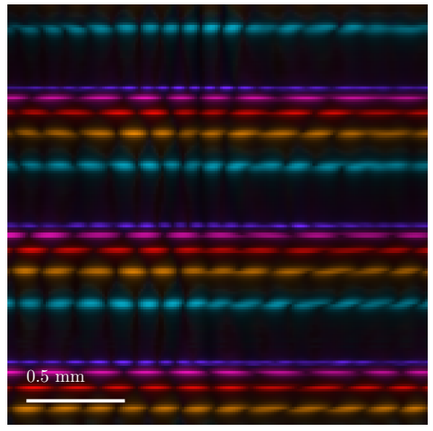
After the physical MLA; all illuminated lenslets.

(4) Full detector



1024 x 1024 sensor; 13 µm pixels. Box marks 4z.

(4z) Detector detail



Exact 164 x 164 pixel crop; same contrast as 4.

Front: native exterior rays. Relay: independent Snell trace continuing those same rays. Vendor internals are not invented. Intensity panels are native POP, with display contrast adjusted. Blue sampling artifacts and broadband performance remain under qualification; see page 2.

400 500 600 700 800

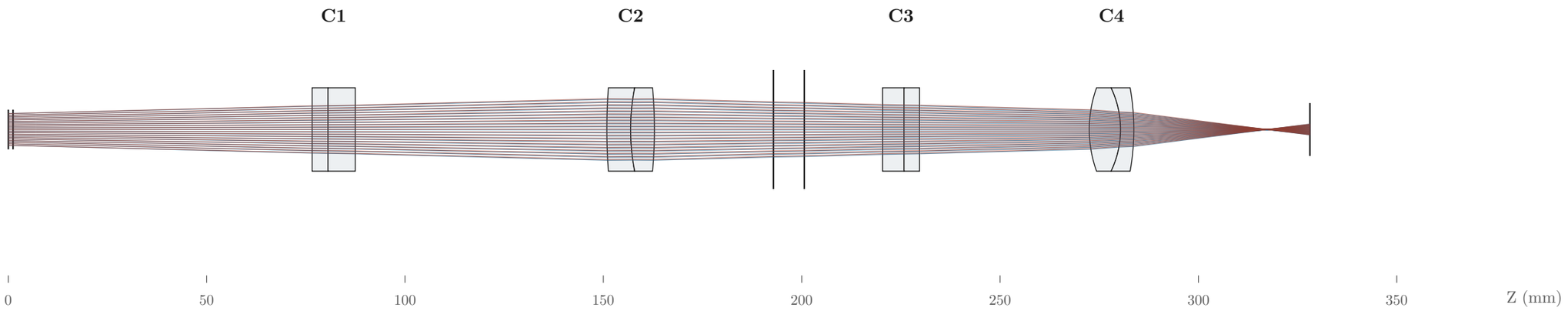
nm; equal-weight samples, not a continuum; 800 nm false-colour.

Mask-to-sensor: 488.1 mm axial. Geometric rays are clean-wavefront references; the intensity panels include turbulence.

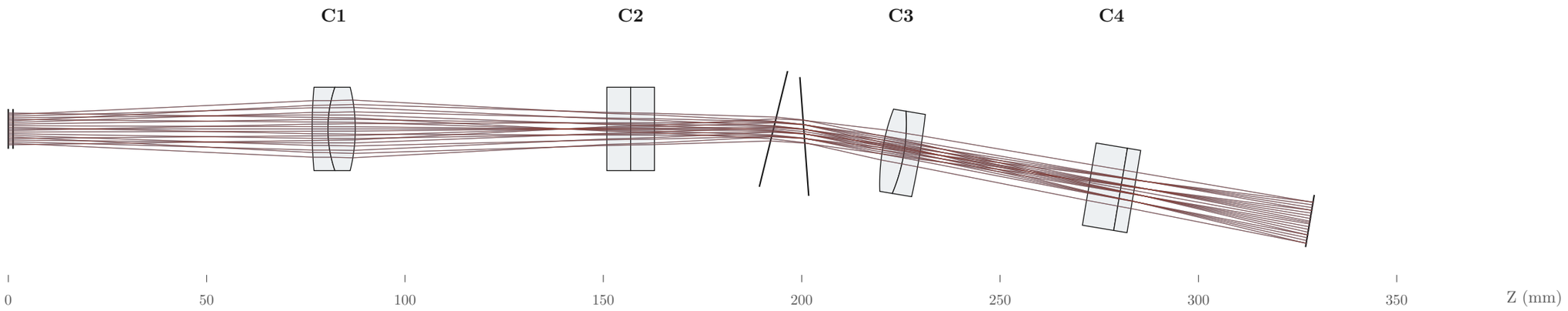
The anamorphic relay: physical powers and scales

Enlarged orthographic sections of the same model; identical X/Z and Y/Z scales. Both sections continue exported arm A/B rays.

Fringe-axis section: X-Z



Lenslet / dispersion-axis section: Y-Z



Nominal magnifications: $|M_X| = 50/150 = 1/3$; $|M_Y| = 100/75 = 4/3$. The relative anamorphism is 4:1. Mask to MLA: 160 mm; MLA to sensor: 328.1 mm axial. These geometrical rays do not replace the complex fields used for the native POP intensity panels.

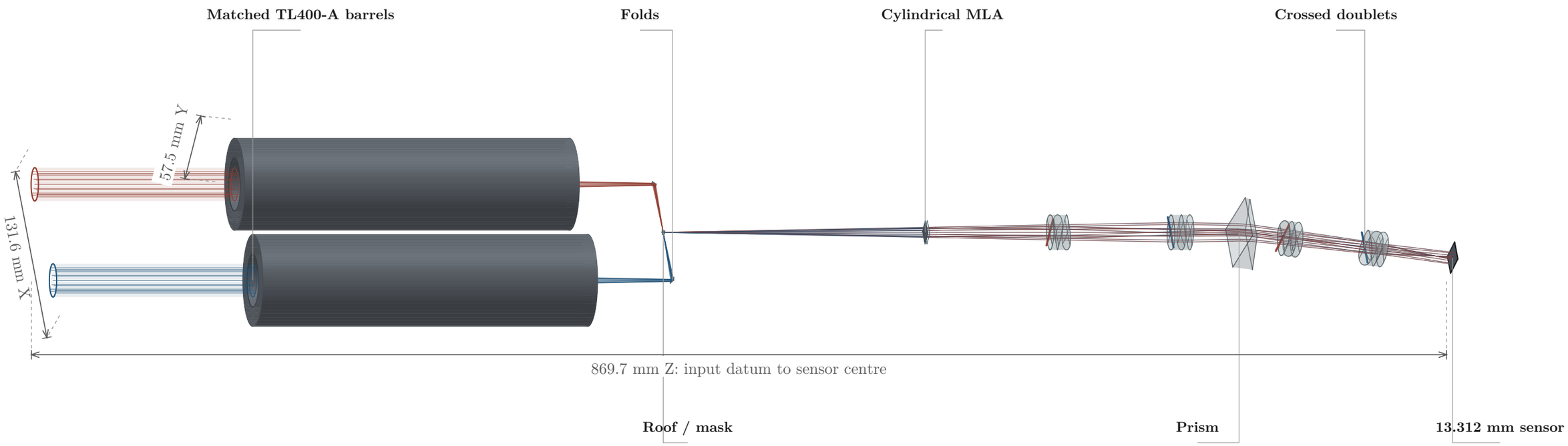
Components per two-beam module

Front pair	2 x Thorlabs TL400-A	f = 400 mm; 20 mm illuminated input per arm
Cylindrical MLA	Edmund 72-590	0.500 mm pitch; R = 3 mm; 1.2 mm fused-silica substrate
C1 / C2 collimators	ACY254-075-A / ACY254-150-A	75 mm Y / 150 mm X; both reversed
C3 / C4 camera	ACY254-100-A / ACY254-050-A	100 mm Y / 50 mm X; four real doublets total
Disperser / detector	18-degree N-BK7 prism / HNu 1024	Prism part TBD; 13 µm pixels; 13.312 mm square sensor

Not yet performance-qualified: P18 tests blue transfer-sampling artifacts; current blue sampling is 7.8 pixels/fringe (goal 2.5). Chromatic focus and spectral packing remain open. TL400-A: 400-700 nm; relay -A coatings: 350-700 nm; 800 nm unqualified. Detector panels use approximate sensor-plane registration and 13 µm bin integration; downstream 1 W normalization is not an absolute throughput prediction.

Real-space layout, with continuous two-arm rays

Metric orthographic 3D views. Optical surfaces and separations are model-derived; no ideal powered elements are substituted.

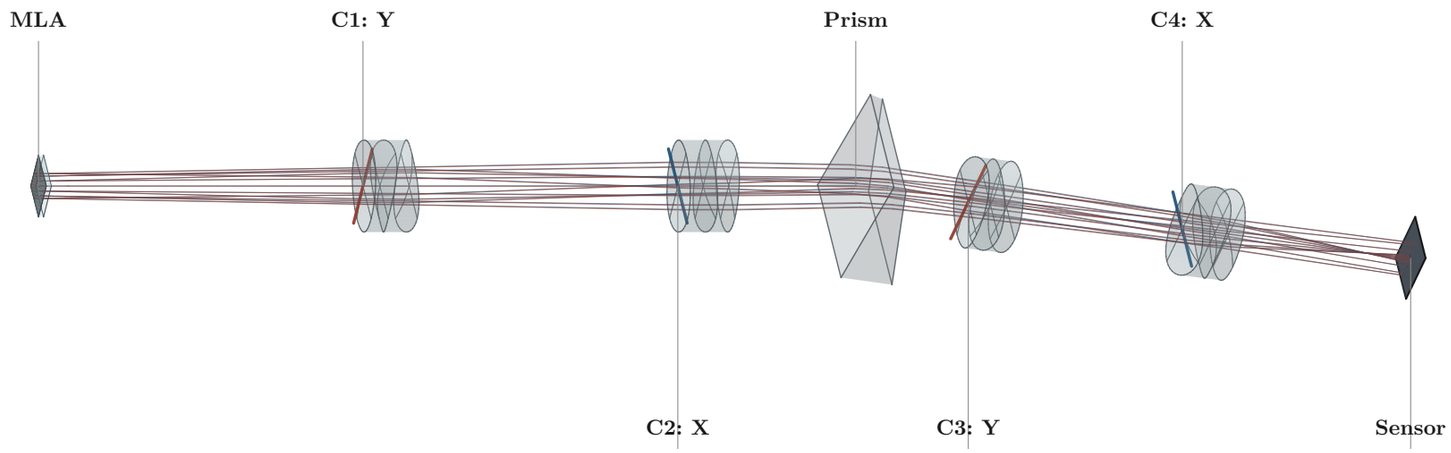


Entrance: end-on X-Y scale check



56.2 mm clear gap; separation / diameter = 3.81

Relay enlargement (uniform magnification)



Scale and scope. Input beams are 20 mm diameter on 76.2 mm centres. The 131.6 mm width includes the two nominal 55.4 mm front barrels; the 57.5 mm height is the drawn optical envelope. Mounts, breadboard, camera housing and access clearances are not included. Relay glass shows 21 mm model clear apertures; the prism has an illustrative 30 mm width. Blue/red rays remain arms A/B throughout both views. Blue/red marks across the doublets denote X/Y power. The end-on view has no foreshortening.